SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2– Case Study (2 QUESTIONS)

Course Code:	ITA05	**Course Title:**	COMPUTER VISION
CO Assessed:	CO3 – Precision (BL4)	**Assessment:**	Assessment Tool 1 – Case Study
Weightage:	40% of CIA	**Total Marks:**	20 (2 Questions × 10Marks)
CO1 Statement:	*Analyze and extract image features using detection and matching techniques for effective image representation and comparison..(BL4)*		
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Section / Batch:	Slot-C	**Date:**	11-08-2026

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Code of Conduct

I _Y . Krishna Tejesh Reddy (192411200)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:

Instructions

- This test contains 2 questions, each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Feature Detection and Matching	Preprocessing, Edge Detection, Harris Corner Detection, Hough Transform, SIFT	1	10
Feature Detection and Matching	PCA, Image Representation, Feature Matching, Image Processing Applications	2	10
		GRAND TOTAL:	**20 Marks**

Question 1: Harris Corner Detection for Satellite Image Registration

The remote sensing system is used to align satellite images of the same geographical region captured at different times. Initially, edge detection is used to find matching regions. However, changes in illumination, orientation, and background details make edge-based matching unreliable. Therefore, Harris Corner Detection is used to obtain stable and distinctive feature points.

Limitations of Edge Detection

Edge detection identifies boundaries where there is a sudden change in intensity. Although edges are useful for identifying object boundaries, they have several limitations in satellite image registration:

1. **Sensitivity to illumination:** Changes in sunlight, shadows, and weather conditions can change edge intensity and position.
2. **Orientation changes:** When the satellite image is captured from a different orientation, the detected edges may not correspond exactly.
3. **Background details:** Roads, buildings, vegetation, and other background structures can create many unnecessary edges.
4. **Poor localization:** An edge represents a line or boundary rather than a specific feature point, making accurate point-to-point matching difficult.
5. **Unstable matching:** The same geographical feature may produce different edge patterns in images captured at different times.
6. **Noise:** Image noise can generate false edges, resulting in incorrect matching.

Advantages of Harris Corner Detection

Harris Corner Detection identifies **corners**, which are points where intensity changes significantly in more than one direction. These points are generally more distinctive and suitable for image registration.

- It provides **well-localized feature points**.
- Corners can be used as specific reference points between two images.
- It reduces the ambiguity that occurs when matching long edges.
- Distinctive corners can be detected from buildings, road intersections, and other structures.
- It provides better feature correspondence between images.
- It improves the accuracy of image alignment even when images have different background details.

Improvement in Image Registration

The Harris detector calculates the change in image intensity around each pixel in different directions. Points with significant intensity variation in both directions are identified as corners. These corners are then matched between the satellite images.

The registration process can be represented as:

Satellite Images → Preprocessing → Harris Corner Detection → Feature Points → Feature Matching → Transformation Estimation → Image Alignment

Thus, Harris Corner Detection provides more stable and distinctive points than simple edge detection. It improves **feature localization, matching accuracy, and robustness**, making it more suitable for satellite image registration under varying imaging conditions.

Conclusion

Replacing edge detection with Harris Corner Detection makes the registration process more reliable. Corners provide precise and distinctive locations that can be matched between images, resulting in more accurate alignment of satellite images captured at different times.

Question 2: Image Representation for Real-Time Object Detection

The autonomous surveillance system captures high-resolution color images for detecting vehicles and pedestrians. Processing the original color images requires more computation and slows down feature extraction. Therefore, grayscale and binary representations are considered to improve processing speed.

Color Image Representation

A color image normally contains three channels: **Red, Green, and Blue (RGB)**. Processing all three channels requires more memory and computational operations.

Advantages:

- Preserves complete color information.
- Useful when object color is important.

Limitations:

- Requires more processing time.
- Requires more memory.
- Feature extraction becomes computationally expensive.
- May not be necessary when object shape and edges are the main requirements.

Grayscale Representation

A grayscale image contains only one intensity channel instead of three color channels. Therefore, it significantly reduces the amount of data that needs to be processed.

Advantages:

- Lower computational complexity.
- Requires less memory.
- Faster image processing.
- Suitable for edge detection.
- Useful for feature extraction based on intensity, shape, and structure.

Grayscale images are particularly useful for algorithms such as **Harris Corner Detection, SIFT, and edge detection**, where color information is not always essential.

Binary Representation

A binary image contains only two intensity values, generally **0 and 1** or **black and white**.

Advantages:

- Very low computational complexity.
- Requires less memory.
- Makes object/background separation simple.
- Useful for segmentation and shape analysis.

Limitations:

- Large amount of original image information is lost.
- Small intensity variations may disappear.
- Feature extraction can become less reliable if thresholding removes important object details.
- It may not work well when the object and background have similar intensity.

Comparison

Representation	Computation	Information	Edge Detection	Feature Extraction
Color	High	Very high	Good	Good but slower
Grayscale	Medium/Low	Moderate	Very good	Very good
Binary	Very low	Low	Good for clear boundaries	Limited

Most Appropriate Representation

For the given real-time surveillance application, **grayscale representation is the most appropriate choice**. It reduces computational complexity while retaining important intensity, shape, and structural information required for edge detection and feature extraction.

The processing pipeline can be:

Color Image → Grayscale Conversion → Preprocessing → Edge/Feature Detection → Feature Extraction → Object Detection

Binary images can be used additionally when clear segmentation is required, but converting directly to binary may cause loss of useful information.

Conclusion

Grayscale provides the best balance between **computational efficiency and detection accuracy**. It reduces processing time compared with color images while preserving sufficient information for reliable edge detection and feature extraction. Therefore, grayscale representation is recommended for accurate and fast real-time vehicle and pedestrian detection.